Software Requirements Specification

Smart Parking System

Structured technical document aligned for formal review, sign-off, and handover.

Document ID	Version	Date	Author
SRS-20260514-131843	1.0	2026-05-14	Ali

Req2Design IEEE 830-1998 aligned Generated draft for expert review

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Revision History

Name	Date	Reason For Changes	Version
Ali	2026-05-14	Initial generated draft for review.	1.0

1: Introduction

1.1: Purpose

Purpose: To define the Smart Parking System's software requirements ensuring seamless operation and user satisfaction.

1.2: Scope

Scope: This Software Requirements Specification (SRS) covers functionalities required for drivers, gate operators, and administrators within the Smart Parking System.

1.3: Definitions/Acronyms

- **SP:** Smart Parking System
- **DR:** Driver
- **GO:** Gate Operator
- **ADM:** Administrator

1.4: Overview

Overview: The SRS outlines how the SP will support real-time viewing, slot reserving, payment processing, and management activities for all stakeholders involved.

2: Overall Description

Perspective: From a system perspective, the SP is designed to facilitate efficient parking through automated processes and real-time information updates.

Functions:

- Real-time display of parking lot status
- Slot reservation and digital payment
- QR code validation and manual override
- Incident reporting
- Reservation expiration handling
- Dynamic pricing adjustment based on demand
- Occupancy monitoring and analytics
- Revenue tracking and maintenance scheduling

Users: Drivers, gate operators, and administrators.

2.4: Constraints

Constraints: Limited network connectivity may affect real-time updates.

2.5: Assumptions/Dependencies

- Reliable internet connection for real-time updates.
- Validity of payment methods accepted.

3: Specific Requirements

External Interfaces:

- Web interface for drivers and administrators
- Mobile app for drivers
- Physical gates interfacing with QR codes

3.2: Functional Requirements

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- Real-time Display of Parking Lot Status

Input: User requests current parking lot status via web/mobile interface.

Processing: System queries connected sensors and displays available slots in real-time.

Output: Updated list of available parking slots shown to the user.

Priority: High

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- Slot Reservation and Digital Payment

Input: Driver selects desired slot and initiates reservation process.

Processing: System verifies slot availability, calculates fees, prompts for payment method, and confirms reservation.

Output: Confirmation message displayed indicating successful reservation and payment.

Priority: High

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- QR Code Validation and Manual Override

Input: GO scans QR code associated with reserved space.

Processing: System validates reservation details against sensor data; if mismatch, manual override allows entry.

Output: Entry granted upon valid reservation confirmation or operator approval.

Priority: Medium

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- Incident Reporting

Input: GO reports incidents such as blocked or damaged spots using mobile app.

Processing: Report is logged and sent to maintenance team for action.

Output: Notification sent to relevant personnel regarding reported issue.

Priority: Low

Non-Functional Requirements

Usability: Interface shall be intuitive and accessible to users across different devices.

Reliability: System must ensure accurate and consistent operation even under varying loads.

Performance: Response times for critical actions (reservation, payment, entry) shall not exceed 5 seconds.

Portability: Platform must run on multiple operating systems and browsers.

4: System Features

Real-time parking lot status update mechanism.

Slot reservation and payment facilitation.

QR-based access control with fallback options.

Incident logging and reporting functionality.

Dynamic pricing algorithm.

Occupancy analysis dashboard.

Revenue summary report generation.

Maintenance schedule planner.

Alert system for overstay violations.

Immutable transaction log storage.

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